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So you thought the megahertz war was over? Dead wrong; the battle for clock speed simply carried on over a different battleground — the multi-core one. In the time before the Core 2 Duo and Athlon 64 Intel and AMD were determined to out-clock each other and with little heed for whose architecture was actually superior they kept slugging it out. Intel was the major culprit of the two; their Pentium 4 CPUs scaled terrific speeds of up to 3.8 GHz and in the bargain provided users not only with a CPU but a free room heater as well. AMD finally used some grey matter and came back with the Athlon 64 range of CPUs. Thanks to brilliant design, greater IPC, (Instructions Per Clock), and a highly efficient memory controller integrated on the CPU die itself these processors enjoyed a peaceful dominion for some two years; and AMDs depleted coffers gradually filled up with the spoils; nobody in the desktop space with any common sense was even buying Intel anymore. These well-gotten gains were used by AMD to gobble up ATi Technologies in the hope of procuring good

Meet the latest bad boys from AMD and Intel. They're fast, they're mean, and they're gunning for a place in your machine

integrated graphics solutions for their platforms and for just a little while AMD was slightly remiss in its duty to develop the best CPUs around, which is obviously a function of cash-in-hand. This is a business where even slim opportunities are capitalised upon, and Intel is nobody's fool. Silent for nearly two years, smarting from the back-handed insult the Athlon 64s had delivered; nestled deep inside their research bunkers in Israel, a team of Intel engineers had prepared a little surprise they codenamed "Conroe"; an architecture that did away with the Pentium 4s *NetBurst Microarchitecture* totally and had a number of new, radical improvements. We know the CPUs based on this architecture as *Core 2 Duo*, a name that has become synonymous with the

term performance over the past two years.

Core 2 Duo methodically destroyed Athlon 64 overnight, and Intel was basking in the glow of enthusiast affection again.

One thing that the Core architecture did was delve deep into multi-core processing territory; from where the CPU industry has yet to return; and perhaps never will. Parallelism is the future and we're seeing dual-core processors fast becoming the *de facto* standard globally. Newer tri-core and quad-core CPUs are also becoming widespread; their main functional principle? Simple, more hands can do more work.

Sadly application support for these tiny computing marvels is sorely lacking; and applications have only just started featuring support for up to two processing cores. A select handful of applications have incorporated support for more than two cores. The reason for this dearth in application support is simply volumes; what's the use of spending resources developing and optimising a software application for a quad-core processor when only some five per cent of the total desktops around the world feature these processors? Such multi-core CPUs spoil

War At The Core



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applications by giving developers the liberty to assume infinite processing resources, which leads to a whole new can of worms. Then there are those applications that benefit more from increased core clocks, a good example being 3D games in general. In fact any application that doesn't support more than a single CPU core will respond to higher clock speeds favourably. Obviously a quad-core CPU cannot maintain the same clock speed as a dual-core CPU; the laws of thermal dissipation just don't allow it. Similarly a single core will be able to main-

tain a higher stable clock speed than a dual-core; and will generate less heat at any given clock speed. Now there's a dilemma — what do you buy, if you were shopping for a processor? Do you look at performance primarily and go for a dual-core, (single cores are history now, so they're out), or do you invest in a quad-core; which surely would be more future proof, as applications will eventually become heavily multithreaded? Then there's the all important factor of cost. Do you spend Rs 10,000 on a 3.2 GHz dual-core or do you opt for a 2.5

GHz quad-core instead? There's no simple answer for this question. Some applications will run better on the quad-core, while others will favour the dual-core, so your choice of application dictates which processor would be better. Obviously you could hunt around for a 3.2 GHz quad-core, but then you would probably pay around Rs 25,000 for it; a hike of 250 per cent, for a performance delta of not more than 30 per cent. We've got a mix of dual-, tri-, and quad-core CPUs in this comparison; ten CPUs from AMD and five from Intel and we've

How We Tested

We used the following components on our test machines:

Test Bench Components	
RAM	2 x 2 GB, Corsair Dominator DDR2 1066 MHz
Video Card	ASUS EN GTX 280 TOP (670 MHz core)s
HDD	WD Raptor 10,000 rpm, 80 GB SATA
OS	Windows XP Prof SP 2

For AMD boards we used the ASUS M3N HT-Deluxe Heat-Pipe Edition motherboard based on NVIDIA's 780a SLI chipset. Our Intel platform was based on an ASUS Rampage Formula motherboard based on Intel's X48 chipset. We use the best components possible to ensure these processors were not bottlenecked in any way. We also used identical test bench rigs for both systems to minimise the influence of components other than the processors. This ensures the most accurate results, since any change in performance can only be attributed to the processor. We did not rate features at all, these are largely unimportant for CPUs, since performance is a better indicator of how good a particular feature is or not. Our benchmarks were divided into two categories — Synthetic and Real World.

PC Mark 05 is a standard benchmark that tests all major subsystems of a PC. We noted only the processor score from the scores received. This score is arrived at by stressing the CPU with many tasks like encoding, extracting, physics, math calculations and simulations. 3D Mark 2006 tests a computer's CPU and graphics card. We noted the CPU score only, this benchmark runs a number of physics, AI, and render routines to arrive at the CPU score. wPrime is a multithreaded benchmark that stresses CPUs by computing the square roots of many numbers at once. We use the 32M setting, which computes the square root of 32 million numbers. The result is returned in seconds taken to complete the task; obviously the lower this value is, the faster the processor. ScienceMark 2.0 is a suite of benchmarks that consists of molecular simulations, encryption algorithms and other benchmarks that stress out your CPU. This benchmark is used for theoretical engineering and scientific computing. MemBench tests the speed of the CPUs L1 and L2 cache and the memory speed; we were interested in the latter. The Cryptography part

of the test uses Rijndael's method of AES encryption and tests your CPUs number crunching ability.

CineBench R10 is CineBench's latest benchmark that tests the CPUs rendering capabilities in rendering a fixed scene with fixed lighting. It is heavily multithreaded and you will see a quad core CPU distributing the rendering workload into four parts at the beginning. If one core finishes its workload early it jumps in and helps another core finish up. CineBench is an important theoretical benchmark for those intending to use their PCs for rendering. POV-Ray 3.7 is an acronym for Persistence Of Vision Raytracer; this is a free tool that can be used to create 3D scenes with light effects. We use an inbuilt scene called Chess 2 and this scene is rendered at a resolution of 800 x 600 pixels. The time taken to do so is duly noted in seconds, the quicker the CPU, the better it is.

For real world rendering tests we use Maya 8.0. We rendered a scene called Globalillum at a resolution of 640 x 480 pixels at production quality settings. This benchmark can be set to use one, two, three or more cores. Once again, the less the time taken, the faster the CPU is. WinRAR 3.8 is the latest compression tool from RAR labs. We use the inbuilt benchmark that provides us a score in kilobytes per second, obviously the higher this value the faster the CPU is at compression/decompression tasks. DivX Encode 6.8.0.19 is a popular tool for DVD ripping and video encoding. We used a 100 MB .VOB file and ripped this to a home theatre grade DivX file. The time taken to complete this is measured, and the fastest CPUs are the ones that do it the quickest.

Lastly we ran three games — *Doom 3*, *Company Of Heroes* and *Unreal Tournament 3*. These games were run at minimum resolutions and detail settings to make them as CPU bound as possible. This is done to reduce the role of the graphics card in the benchmark, obviously the role of the GPU cannot be eliminated, but it can be minimised.

Pre-setup we flashed the motherboards to their latest stable BIOS'. The latest drivers were downloaded from their respective sites. For the GTX 280 we used NVIDIA's drivers V 178.13. After installing all benchmarks, disk defragmenter was run to ensure minimal fragmentation. Each of the benchmarks was run thrice and a mean figure taken.